

Midterm Solution
Nov 5th

1. (20 points) Consider a discrete time Markov chain (DTMC) $\{X_n, n \geq 0\}$ on the state space $E = \{0, 1, 2\}$ with transition probability matrix given by

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 1/3 & 1/3 & 1/3 \end{pmatrix}$$

Assume that the (initial) distribution of X_0 is the uniform distribution on E .

- (a) (5 points) Calculate $\mathbb{P}(X_2 = 2)$.
 (b) (10 points) Calculate $\mathbb{P}(X_4 = 2, X_3 \neq 0 | X_1 = 1)$.
 (c) (5 points) Is there a limiting distribution for this DTMC? Prove your claims.

solution

- (a) First, we calculate P^2 as

$$P^2 = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 1/6 & 2/3 & 1/6 \\ 5/18 & 4/9 & 5/18 \end{pmatrix} \quad (2 \text{ points})$$

$$\begin{aligned} \mathbb{P}(X_2 = 2) &= \sum_{i=0}^2 \mathbb{P}(X_2 = 2 | X_0 = i) \mathbb{P}(X_0 = i) & (2 \text{ points}) \\ &= \frac{1}{3} \sum_{i=0}^2 \mathbb{P}(X_2 = 2 | X_0 = i) \\ &= \frac{1}{3} \sum_{i=0}^2 P_{i,2}^2 \\ &= \frac{1}{3} \left[\frac{1}{2} + \frac{1}{6} + \frac{5}{18} \right] \\ &= \frac{17}{54} & (1 \text{ points}) \end{aligned}$$

- (b)

$$\begin{aligned} \mathbb{P}(X_4 = 2, X_3 \neq 0 | X_1 = 1) &= \mathbb{P}(X_4 = 2, X_3 = 1 | X_1 = 1) + \mathbb{P}(X_4 = 2, X_3 = 2 | X_1 = 1) & (3 \text{ points}) \\ &= \mathbb{P}(X_4 = 2 | X_3 = 1, X_1 = 1) \mathbb{P}(X_3 = 1 | X_1 = 1) \\ &\quad + \mathbb{P}(X_4 = 2 | X_3 = 2, X_1 = 1) \mathbb{P}(X_3 = 2 | X_1 = 1) \\ &= P_{1,2} P_{1,1}^2 + P_{2,2} P_{1,2}^2 & (3 \text{ points}) \\ &= \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{6} & (3 \text{ points}) \\ &= \frac{7}{18} & (1 \text{ points}) \end{aligned}$$

- (c) Since $p_{22} > 0$, state 2 has period 1 and is thus aperiodic. Moreover, all states communicate with each other, hence the Markov chain is aperiodic. (2 points) Limiting probability exists since the MC is finite, irreducible and aperiodic. (3 points)

Remark. You need to prove the "aperiodicity" instead of claim it directly.

2. (30 points) Let $N > 3$ be a positive integer. Consider a Markov chain with states $\{1, \dots, N\}$ and transition probabilities (non null ones) given by

- for $1 \leq i \leq (N-1)$, $P(i, i) = 1 - i/N$ and $P(i, i+1) = i/N$;
- $P(N, N-1) = 1$.

- (5 points) Find the communication classes of the chain.
- (5 points) Find the period of states 1 and $N-1$.
- (5 points) Find the transient and recurrent states.
- (5 points) Find a stationary distribution of the chain.
- (5 points) Consider two states i and j where $1 \leq i < j \leq N-2$. Assume that the chain start at the state j . Find the expected amount of time the chain is in the state i .
- (5 points) Consider two states i and j where $1 \leq i < j \leq N-2$. Assume that the chain start at the state i . Find the expected amount of time the chain is in the state j .

solution:

- $\{N-1, N\}$ (2.5 points);
 $\{i\}$ for $i = 1, \dots, N-2$ (2.5 points).

Remark. $i \Leftrightarrow i$ and therefore i communicates with itself when $i \leq N-2$ and the communication class $\{i\}$ is a singleton

- $d(N-1) = d(1) = 1$ since $p_{ii}^1 > 0$ for $i = 1, N-1$. (5 points)

- Transient: $\{1, \dots, N-2\}$ (2.5 points);
 Recurrent: $\{N-1, N\}$ (2.5 points).

- Clearly $\pi_i = 0$ for all transient states i . (1 points)

Further we have $\pi = (\pi_{N-1}, \pi_N)$ such that

$$\pi_{N-1} + \pi_N = 1 \quad (1 \text{ points})$$

$$\pi_{N-1}(1 - \frac{N-1}{N}) + \pi_N = \pi_{N-1} \quad (2 \text{ points})$$

Solving for the system of equations yields that $\pi_{N-1} = \frac{N}{2N-1}$, $\pi_N = \frac{N-1}{2N-1}$. (1 points)

- 0 since $i \leq N-2$ cannot come back to $j < i$. (5 points)

Remark. It is wrong to say that the expected time spent in j is ∞ , implying that it stays in j for infinitely many times. Notwithstanding, we still give you 3 points when grading this problem.

- Denote the expected time as s_{ij} .

$$s_{ij} = (1 - \frac{i}{N})s_{ij} + \frac{i}{N}s_{i+1,j} \text{ if } i < j \quad (2 \text{ points})$$

$$s_{jj} = 1 + (1 - \frac{j}{N})s_{jj} \quad (2 \text{ points})$$

which yields $s_{ij} = s_{i+1,j} = \dots = s_{jj} = \frac{N}{j}$. (1 points)

Remark. When you take one-step analysis, there is no "1" in the left-hand side for s_{ij} because when you move from i to the next point, the time stayed in j does not increase. See Chapter 1.7 in the lecture notes for more elaborations.

3. (30 points) Travelers visit the gift store at a Poisson rate of 10 per minute. Suppose that 20 percent of travelers who visit the store are Chinese. Suppose Amy arrives at this point at 10:00am and observes no travellers at that moment.

- (6 points) What is the probability that at least one traveler visits the store in 1 minute? What's the probability that at least one Chinese traveler visit the store in 3 minutes?

- (b) (6 points) What is the mean time until the 5th traveler visits the store? What is the mean time until the 10th Chinese traveler visit the store?
- (c) (6 points) Given that 5 Chinese travelers have visited the store before 10:05am, what is the expected number of travelers who visit the store in that time?
- (d) (6 points) Given that 3 Chinese travelers have visited the store before 10:05am, what is the (conditional) probability that by 10:02am, only one traveler visit the store?
- (e) (6 points) Given that 5 Chinese travelers have visited the store before 10:05am, what is the expected time until the 6th Chinese traveler visit the store?

(remark: it is ok to leave expressions like e^x in your answers).

Solution.

- (a) Let X be the number of travelers having visited the store in 1 minute. Then, we know that

$$X \sim P(10)$$

Hence,

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - \frac{10^0}{0!}e^{-10} = 1 - e^{-10}. \quad (3 \text{ points})$$

Let Y be the number of Chinese traveler having visited the store in 3 minutes, then

$$Y \sim \mathbb{P}(10 \times 3 \times 20\%) \quad i.e. \quad Y \sim \mathbb{P}(6)$$

Consequently,

$$\mathbb{P}(Y \geq 1) = 1 - \mathbb{P}(Y = 0) = 1 - \frac{6^0}{0!}e^{-6} = 1 - e^{-6}. \quad (3 \text{ points})$$

- (b) Let T be the total time until the 5th traveler visits the store, T_i for $i = 1, \dots, 5$ be the interarrival time for the first 5 travelers. Then, we have

$$\mathbb{E}[T] = \mathbb{E}[T_1 + T_2 + T_3 + T_4 + T_5] = 5\mathbb{E}T_1 = \frac{1}{2}. \quad (3 \text{ points})$$

Let T^1 be the total time until the 10th traveler visits the store, T_i^1 for $i = 1, \dots, 10$ be the interarrival time for the first 10 Chinese travelers. Then, we have

$$\mathbb{E}[T^1] = \mathbb{E}\left[\sum_{i=1}^{10} T_i^1\right] = 10\mathbb{E}T_1^1 = 5. \quad (3 \text{ points})$$

- (c) Let X be the number of travelers who visit the store in that time, Y be the number of Chinese travelers visiting the store before 10 : 05am, and Z be the number of non-Chinese travelers visiting the store before 10 : 05am.

By the coloring theorem, we know that $Y \sim \mathbb{P}(10)$ and $Z \sim \mathbb{P}(40)$ are independent. (3 points)

Hence, we have

$$\mathbb{E}[X|Y = 5] = \mathbb{E}[5 + Z] = 5 + \mathbb{E}Z = 45 \quad (3 \text{ points})$$

- (d) Let N_t be the number of travelers visiting the store before t , N_t^1 be the number of Chinese travelers visiting the store before t , and N_t^2 be the number of non-Chinese travelers visiting the store before

t. Based on the fact $\{N_2 = 1\} = \{N_2^1 = 0, N_2^2 = 1\} \cup \{N_2^1 = 1, N_2^2 = 0\}$, we have

$$\begin{aligned}
\mathbb{P}(N_2 = 1 | N_5^1 = 3) &= \frac{\mathbb{P}(N_5^1 = 3, N_2 = 1)}{\mathbb{P}(N_5^1 = 3)} \\
&= \frac{\mathbb{P}(N_5^1 = 3, N_2^1 = 1, N_2^2 = 0) + \mathbb{P}(N_5^1 = 3, N_2^1 = 0, N_2^2 = 1)}{\mathbb{P}(N_5^1 = 3)} \quad (3 \text{ points}) \\
&= \frac{\mathbb{P}(N_5^1 - N_2^1 = 2, N_2^1 = 1, N_2^2 = 0) + \mathbb{P}(N_5^1 - N_2^1 = 2, N_2^1 = 0, N_2^2 = 1)}{\mathbb{P}(N_5^1 = 3)} \\
&= \frac{\mathbb{P}(N_5^1 - N_2^1 = 2)\mathbb{P}(N_2^1 = 1)\mathbb{P}(N_2^2 = 0) + \mathbb{P}(N_5^1 - N_2^1 = 3)\mathbb{P}(N_2^1 = 0)\mathbb{P}(N_2^2 = 1)}{\mathbb{P}(N_5^1 = 3)} \quad (2 \text{ points}) \\
&= \frac{\frac{6^2}{2!}e^{-6}\frac{4^1}{1!}e^{-4}e^{-16} + \frac{6^3}{3!}e^{-6}e^{-4}\frac{16^1}{1!}e^{-16}}{\frac{10^3}{3!}e^{-10}} \\
&= \frac{486}{125}e^{-16} \quad (1 \text{ points})
\end{aligned}$$

Remark. The original solution has a mistake. The correct way to divide the event $N_2 = 1$ is that "1 Chinese come and 0 non-Chinese come before 2 minutes" and "0 Chinese come and 1 non-Chinese come before 2 minutes". Mathematically,

$$\{N_2 = 1\} = \{N_2^1 = 1, N_2^2 = 0\} \cup \{N_2^1 = 0, N_2^2 = 1\}$$

where the LHS is two disjoint events. And be noted that $\{N_2 = 1\} \neq \{N_2^1 = 1\} \cup \{N_2^1 = 0\}$! Given this concern, we have re-graded this problem and adjust your scores for fairness.

- (e) Let T be the arrival time of 6th Chinese traveler visiting the store and T' be the first arrival time of Chinese traveler from $t = 10 : 05\text{am}$, then we know that $T' \sim P(2)$. Hence, we have

$$\mathbb{E}T = \mathbb{E}[5 + T'] = 5 + \mathbb{E}T' = 5 + \frac{1}{2} = \frac{11}{2}. \quad (6 \text{ points})$$

Remark. We relax this grading criteria as follows. If you have argued that T' follows the exponential distribution with parameter 2, you earn 4 scores. Correctly computing the expectation $\frac{1}{2}$ and $\frac{11}{2}$ earns 2 more scores.

In addition, the problem implies that the 6th Chinese traveler comes after 5 minutes so you can simply use memory-less property of exponential distribution.

4. (20 points) Suppose Amy plays a game, where in each round she succeeds with probability $p \in [0, 1]$ and loses with probability $q = 1 - p$ respectively.
- (a) (5 points) Suppose the game ends if Amy has two consecutive successes or has two consecutive losses. Let X be the number of rounds, what is the distribution of X ?
- (b) (5 points) Let C_1 be the number of rounds required until Amy has 2 *consecutive* successes, and C_2 be the number of rounds required until Amy has 2 *consecutive* loses. What is the expected value of C_i ($i = 1, 2$)?

Now suppose that rounds of game are performed at the successive arrival times of a Poisson process with rate $\lambda = 2$. Let T_1 be the time waited for *totally* having 3 successes, and T_2 be the time waited for *totally* having 3 loses.

- (c) (5 points) Are T_1 and T_2 independent? Justify your answer.
- (d) (5 points) Find $\mathbb{P}(T_i > t)$ for $t > 0$ and $i = 1, 2$.

Solution.

- (a) The distribution of X is given as

$$\mathbb{P}(X = k) = \begin{cases} (pq)^n & k = 2n + 1 \text{ for } n \geq 1 \\ (pq)^{n-1}(p^2 + q^2) & k = 2n \text{ for } n \geq 1 \end{cases}$$

5 points for the answer. If students only consider special cases such as $p = 1/2$, then they earn at most 2 points.

- (b) We use T_0 to denote the expected number of rounds to have 2 consecutive successes, starting from 0 success. We use T_1 to denote the expected number of rounds to have 2 consecutive successes, starting from 1 success. We then have the following recursive formulas:

$$\begin{cases} T_0 = p \cdot (T_1 + 1) + q \cdot (T_0 + 1) \\ T_1 = p \cdot 1 + q \cdot (T_0 + 1) \end{cases}$$

Solving for the above two equations, we have $T_0 = \frac{1+p}{p^2}$ and this is equivalent to C_1 . Due to symmetry, we have $C_2 = \frac{1+q}{q^2}$.

3 points for the recursive formula and 2 points for the answers.

Remark.

- i. Please see in Exercise 5, Tutorial 2 for how to construct the Markov chain for such successive wins or losses. Please also see in Exercise 4, Tutorial 4 for how to calculate the mean time by one-step argument.
 - ii. We do not encourage you to use series to solve this kind of problem. Frankly speaking, it creates much more trouble to correctly enumerate the events which exhaust all the possibilities.
 - iii. Also, the one-step method is emphasized not only in lecture notes, but also in Tutorial/Quiz/Homework. This method also take merits especially when the chain becomes complicated (but you only need to solve the system of linear equations).
 - iv. See the famous gambler's ruin problem for more elaborations if you are not familiar with this method.
- (c) Call arrivals where head or tail appears type- i arrivals, $i = 1, 2$. According to the decomposition theorem of Poisson processes, the two type of arrivals form two *independent* Poisson processes, say $\{N_t^i\}_{t \geq 0}$, with respective rate parameters $\lambda_1 = 2p$ and $\lambda_2 = 2q$. Thus $N_t = N_t^1 + N_t^2$ is the initial (combined) Poisson process. As the T_i s are the waiting times for the 3th arrival of type i , they are independent.

3 points for the coloring theorem, 2 points for the remaining argument.

- (d) We have for $t > 0$,

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N_t^1 < 3) = e^{-2pt} \sum_{k=0}^2 \frac{(2pt)^k}{k!} = e^{-2pt}(1 + 2pt + 2p^2t^2),$$

$$\mathbb{P}(T_2 > t) = \mathbb{P}(N_t^2 < 3) = e^{-2qt} \sum_{k=0}^2 \frac{(2qt)^k}{k!} = e^{-2qt}(1 + 2qt + 2q^2t^2).$$

3 points for the transformation from T_i to N_t^i , 2 points for the answers.